

Hierarchical Robustness-Aware Weakness Profiling

Key Findings & Innovation Summary

▢ MAIN INNOVATION

Aggregate DOVE robustness scores hierarchically (by MMLU subject)
Transform 2,367 scattered questions → 18 actionable domain insights

Mathematical Clustering

42.2% robustness gap
87.5% weakness rate

Methodology Validation

88% threshold consistency
Balanced 0.5 threshold

Pattern Discovery

Systematic vulnerabilities
invisible at question level

▢ WHY ROBUSTNESS > ACCURACY

ACCURACY:

- Binary right/wrong
- "Model correct on calculus"

DOVE ROBUSTNESS:

- Consistency across formats
- "Fails A,B,C,D → 1,2,3,4"
- Reveals brittle patterns

IMPACT: Enables systematic domain-targeted improvement vs scattered question fixes